Om Patel

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Education

Rutgers University–New Brunswick Sep 2025 – May 2027
Master of Science in Computer Science; GPA: 3.92/4.0 *New Brunswick, NJ*

Pandit Deendayal Energy University Jun 2021 – Jun 2025
Bachelor of Technology in Information & Communication Technology; CGPA: 8.93/10 *Gandhinagar, India*

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, R
Full-Stack: React, Next.js, Node.js, FastAPI, Flask, Django, Prisma, SQLAlchemy, REST APIs
Databases: PostgreSQL, Redis, MySQL, MongoDB
AI/ML: LLM Integration (Llama 3.3 via Groq), Prompt Engineering, PyTorch, TensorFlow, Scikit-learn
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, GitHub Actions (CI/CD), Vercel, Render, Git, Linux, Bash, Locust
Data & Analytics: Pandas, NumPy, Data Visualization, Dashboards, Qualtrics, Excel, Technical Writing

Experience

Research & Assessment Graduate Coordinator (Part-Time) Jul 2026 – Present
Rutgers University – Career Exploration and Success *New Brunswick, NJ*

- Partner with department leadership to scope and iterate on an employer-facing dashboard that shows recruiters career fair attendance by major to attract new employers (Python, HTML/CSS, JSON); used by 10+ staff ahead of public launch
- Replaced manual mentor matching with a rule-based algorithm that produced ∼40 mentor–mentee pairs
- Map student engagement across majors and present charts and written summaries that guide staff outreach decisions

Research Intern – Deep Learning Jan 2025 – May 2025
Pandit Deendayal Energy University *Gandhinagar, India*

- Engineered a Dual-Stream U-Net for satellite change detection, reaching 89% IoU and a 17% higher deforestation-detection F1
- Outperformed FCN-8 baselines on land-cover reconstruction with optimized U-Nets, raising PSNR by 22% (0.85 SSIM)
- Automated preprocessing pipelines for 10,000+ geospatial images (TensorFlow, PyTorch), cutting latency by 40%

Embedded Systems Intern May 2024 – Jul 2024
Candore Bioscience *Ahmedabad, India*

- Developed a system that estimates cryogenic liquid levels from noisy thermal and ultrasonic sensors (ESP32 MicroPython, Arduino Embedded C), improving detection reliability by 40%
- Applied Steinhart-Hart regression and exponential smoothing to temperature signals, enabling stable readings to −23°C
- Delivered real-time data acquisition for sensor calibration, signal filtering, and anomaly-resistant measurement

Projects

Composur | *Next.js, TypeScript, Llama 3.3 (Groq), PostgreSQL, AWS S3* composur.tech

- Designed and launched a live voice-based technical interview platform end to end, where Llama 3.3 70B (via Groq) asks adaptive questions and returns detailed performance evaluations
- Integrated third-party APIs into a real-time voice pipeline with role-specific prompts for 28 interview domains (Groq LLM inference, ElevenLabs text-to-speech, speech recognition, AWS S3)
- Architected a layered TypeScript backend with session auth and rate limiting (Prisma/PostgreSQL), shipped on Vercel behind GitHub Actions CI with unit and end-to-end tests

EventPass | *Python, FastAPI, PostgreSQL, Redis, Docker, Locust* github.com/Ompatel179/Eventpass

- Designed and shipped a QR-code check-in product for student clubs end to end: sign-up site, QR tickets, organizer dashboard with live counts and CSV export, and a phone scanner, all on one FastAPI + PostgreSQL API
- Guaranteed exactly-once check-in with a database unique constraint: 0 duplicates in 2,862 concurrent scans
- Found and fixed a deadlock with Locust load tests (13 → ∼950 req/s, 0 errors), deployed with Docker on Render and Neon behind GitHub Actions CI (82 tests, 97% coverage)

SRGAN Image Super-Resolution | *Python, PyTorch, GANs, VGG19, OpenCV*

- Implemented a Super-Resolution GAN in PyTorch for 4× upscaling with custom data loading and checkpointing
- Trained the model in two stages (MSE pre-training, then adversarial fine-tuning with VGG19 perceptual loss)
- Integrated PSNR and LPIPS evaluation to measure pixel-level fidelity and perceptual similarity to ground truth

Publications

Harnessing Generative Adversarial Models to Enhance Satellite Imagery Quality Aug 2025

- Published at the 2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA): SRGAN/ESRGAN framework for satellite image reconstruction achieving 34.55 dB PSNR and 0.08 LPIPS